

PUNJAB RETROFITTING BRIDGES

Sector: Infrastructure
Hazard addressed: Flood
Targeted assets: Bridges



Description: The intervention focuses on protection of bridge abutments and implementation of scour protection measures, reducing the structural vulnerability of bridges to flood-induced erosion and hydraulic forces. By securing bridge foundations against scouring and reinforcing abutments, the measure safeguards critical transport infrastructure, maintains connectivity, and reduces flood-related damages to the bridge asset base. However, it needs to be noted that though significantly improving the resilience of bridges, these measures alone are not sufficient to fully improve bridge resilience because bridge decks still remain vulnerable to overtopping and uplift damages.

Priority Districts: Mianwali, Chakwal, Gujrat, Gujranwala, Bahawalpur, Rahim Yar Khan, Lahore

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Per bridge (number of bridges).

Scale assumptions: Bridges located in the target districts.

Key design parameters: Protection of bridge abutments; scour protection i.e. only those retrofit measures which are implementable for existing bridges.

Time horizon: 2030–2035

IMPACT

Risk reduction level: Very High Effectiveness – large reduction in damages (Expert Rank 1).

Time to effectiveness: Effectiveness is realized progressively as individual bridge retrofitting works are completed during 2026–2035, with full effectiveness from 2036 onward.

Expected impact: 50% reduction in mean damage degree (MDD) for bridges under flood events up to the 1-in-100-year return period.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Structural and hydraulic engineering assessments of individual bridges; scour and hydrological analysis per bridge; asset registry and condition survey of bridges in target districts; procurement of qualified civil engineering contractors.

Required institutions: Punjab Communications & Works (C&W) Department (primary implementer); National Highway Authority (NHA) for national road bridges; District governments for local-level facilitation.

Barriers: Limited bridge asset registry and condition data for prioritization; procurement complexity for specialized scour protection and abutment works; budget availability and release timelines; access constraints for bridges in remote or seasonally flooded areas.

COST ASSUMPTIONS

Unit cost: USD 56,554 per bridge.

Capital Expenditure (CAPEX):
USD 1,979,409

Operational Expenditure (OPEX):
USD 566 per bridge per year (for the maintenance of the retrofits only).

CO-BENEFITS & TRADE-OFFS

Environmental: Scour protection stabilizes riverbed morphology near bridge foundations, reducing downstream erosion; reduced risk of bridge collapse limits debris and construction material entering waterways; maintained infrastructure reduces need for emergency reconstruction using resource-intensive methods.

Social: Improved connectivity and maintained access routes for communities during and after flood events; reduced risk to lives from bridge collapses under flood conditions; continuity of emergency response, evacuation, and relief supply routes in flood-affected districts.

Economic: Maintained supply chains, market access, and freight movement; reduced indirect economic losses from disrupted transport connectivity in agricultural and industrial districts.

Potential risks: Traffic disruption and detours during retrofitting construction; physical access constraints for retrofitting of bridges in remote or flood-prone areas; residual risk for flood events exceeding the 1-in-100-year design threshold; residual risks because of no retrofit interventions for bridge deck protection; works on bridges over active rivers pose occupational health and safety challenges.

BENEFICIARIES

Primary beneficiaries: Local communities dependent on bridge connectivity, emergency response and relief agencies, and commercial freight operators.

Distributional aspects: Likely to be particularly beneficial for rural and peri-urban communities in flood-prone areas with limited alternative river crossing points, and for populations in districts with high flood exposure.